M.Tech Project Report

Adaptive Quantum-Safe Security and Privacy for Healthcare Data

System: **Agile-HQ-PQC** Controller: **CADA-AASM**

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1 Project Title

Adaptive Quantum-Safe Security and Privacy for Healthcare Data

Item	Detail
System name	Agile-HQ-PQC
Controller	CADA-AASM (Context-Aware Dynamic Agility and Adaptive Anonymization Steering Matrix)
Programme	M.Tech (CSE, Cybersecurity)
Student	Jose Rohit M
Guide	Dr. Karthikeyan Jayaraman
Institution	School of Computer Science Engineering and Information Systems (SCORE), VIT Vellore
Nature of work	Software-only; open implementation with a React operator console, an automated test suite and a Linux network-namespace testbed

Table 1: Project summary.

The title carries two ideas. *Quantum-safe* means that transport protection is built from the post-quantum algorithms that NIST has standardised, ML-KEM for key establishment [1] and ML-DSA for signatures [2], used together with classical algorithms in a hybrid. *Adaptive* means that the strength of that protection, and the strength of the privacy transformations applied to the data, are chosen at run time from measured network conditions rather than fixed at deployment. The two are controlled together, by one controller, under one latency budget.

2 Problem Statement

A hospital must protect patient data against future quantum attacks and must also de-identify it for research. Stronger protection costs time that real-time patient monitoring does not have.

2.1 Why protection is now urgent

Clinical data such as genomes remains sensitive for decades. An adversary can therefore record encrypted traffic today and decrypt it once a sufficiently capable quantum computer exists ("harvest now, decrypt later"). Because the confidentiality lifetime of the data exceeds the expected arrival date of such a machine, waiting for the threat to materialise is not an option. NIST has standardised the replacements: ML-KEM for key establishment [1] and ML-DSA for signatures [2], and its transition guidance on cryptographic agility recommends that organisations be able to change algorithms without re-engineering their systems [3].

2.2 Why it is costly

The new objects are far larger than the classical ones they replace. An ML-DSA-87 signature is 4,627 bytes against 64 bytes for Ed25519, and an ML-KEM-1024 ciphertext is 1,568 bytes against 32 bytes for X25519. A signed ICU telemetry packet therefore spans several Ethernet frames, which hurts most when the network is congested. Streaming telemetry tolerates only

about 200 ms end to end. The measured literature confirms that the cost is real but not uniform: a hybrid ML-KEM-768 plus ML-DSA-65 TLS handshake for MQTT took 212 ms against 44 ms classical on an emulated ARM64 link [7], whereas an ML-KEM-512 handshake on a Raspberry Pi 3 took 14.1 ms [12]. The penalty depends on the platform, the parameter set and the link, so a single fixed choice is bound to be wrong somewhere.

2.3 Why it is not solved

Data released for research must also be de-identified, using differential privacy for signals and (K, L)-anonymity for records. Today, transport security and data privacy are configured separately, and neither reacts to live network conditions. A fixed maximum-strength setting risks missing the 200 ms deadline on weak links, while a fixed weak setting leaves patients exposed. The surveyed work that adapts post-quantum settings at run time does so for a single device [6] or in a drone-swarm simulation [10], and the surveyed adaptive-privacy work reacts to model training rather than to the network [17, 18, 21].

2.4 The problem to solve

> *How can a healthcare gateway choose, at run time and without human intervention, the strongest combination of post-quantum transport protection and privacy parameters that still meets the latency budget, while never lowering the protection of raw patient data below a safe minimum?*

The statement contains three requirements that are in tension. **Strongest**: the system should not be conservative when the link is healthy. **Meets the latency budget**: the system should not be aggressive when the link is stressed. **Never below a safe minimum**: the second requirement must never be satisfied by weakening the protection of raw clinical data. The project is about satisfying all three together.

3 Abstract

This project treats quantum-safe transport security and data privacy as one latency-constrained trade-off, and builds a gateway that picks the strongest protection the network can currently afford.

A controller, CADA-AASM, computes a live latency budget $L = 200 - \mathrm{RTT} - 0.5 \cdot \mathrm{CPU}$ and maps it to one of three tiers. Each tier fixes together a post-quantum key-encapsulation parameter set, a signature parameter set, a data cipher, a differential-privacy parameter ε and anonymity parameters K and L. A three-cycle persistence filter prevents cipher flapping, and raw clinical telemetry is held at NIST Category 3 or above whatever the congestion. Sessions are re-keyed make-before-break, and post-quantum signatures are batched per epoch.

In 30 paired trials under an emulated congestion model, steering cut SLA violations during stress against static Tier 5 (clinical 7.4% to 4.1%, research 6.9% to 2.9%, $p < 10^{-4}$) and cut tier changes by 65%. A real-link validation, using Linux network namespaces, kernel bandwidth shaping and real TCP, did not reproduce that gain at 2 Mbit/s and above. It showed that adaptation matters only on very thin links: at 0.5 Mbit/s static Tier 5 failed on every cycle, adaptive on 13%, and static Tier 3 on none. The distinctive contribution is therefore the joint, safe adaptation of

cryptographic and privacy parameters under a compliance floor, not a latency win. The system is open, software-only and covered by 65 automated tests, with a React operator console.

4 Objectives

The project has one aim: to keep quantum-safe protection and privacy as strong as the network allows, without ever dropping below a safe floor for raw patient data. It is broken into seven objectives.

1. **Joint model.** Define three protection levels (High, Medium, Low; Tier 5, 3, 1) that bind the key-encapsulation set, signature set, data cipher, differential-privacy ε and anonymity (K, L) together, so that one decision configures the whole stack.
2. **Closed-loop controller.** Build CADA-AASM, which turns a measured latency budget into a tier, with a persistence filter against cipher flapping.
3. **Safety floor.** Hold raw clinical telemetry at NIST Category 3 or above under any congestion, and enforce $K \geq 5$, $L \geq 3$ with independent verification on every research export.
4. **Secure channel.** Implement a hybrid X25519 + ML-KEM handshake, make-before-break re-keying, and per-packet Ed25519 plus per-epoch ML-DSA signatures.
5. **Governance.** Provide role-based access (admin, auditor, researcher, operator), a tamper lab and a tamper-evident audit log.
6. **Honest evaluation.** Measure against static baselines with paired statistics, an ablation and a bandwidth sweep, then validate on a real shaped network link and measure native cryptographic cost.
7. **Usable artefact.** Deliver an open, tested implementation with a React operator console and a guided heavy-load demonstration.

Success criteria. (i) The clinical floor holds in every trial. (ii) The evaluation states plainly where adaptation helps and where it does not. (iii) All 65 automated tests pass.

5 Literature Survey

5.1 Scope and method

The survey reads 18 papers from IEEE Xplore, all published between 2023 and 2026, together with the NIST guidance [3]. Four IEEE Xplore searches covered post-quantum cryptography in healthcare, adaptive crypto-agility, differential privacy for medical data, and post-quantum TLS performance. About 20 candidates came back and 18 were kept for relevance. The full text of each of the 18 papers was then read, so the tables below report what each paper does and what it measured, not what its title suggests.

Limits of the method. The search was done by a single reviewer on IEEE Xplore only, so this is not a systematic review. The research gap in Section 5.5 holds for these 18 papers and for the seven non-IEEE works in Section 8.3, not for all published work. Reference [25] is a student project and has not been peer reviewed.

5.2 What each paper does

Table 2: Survey of the 18 IEEE Xplore papers.

Ref	Paper (venue, year)	What it does	Result or limit that matters here
[4]	El-Hadedy et al., DCAS 2024	Kyber and ASCON on a Kubernetes (K3S) cluster of small ARM boards for IoMT	Kyber512 encapsulation took 3,464 μs. Power rose from 4.49 W to 32.5 W. The algorithm is chosen once per deployment.
[5]	Yadav et al., IEEE AIC 2023	Ring-LWE and PUF authenticated key exchange for IoMT, assisted by blockchain and fog nodes	Security analysis and computation/communication cost analysis. No load or latency adaptation.
[6]	Sinaga et al., ICCIT 2025	Rule-based tuning of Kyber and Saber parameters from CPU and RAM load	Light variants (Kyber512, Saber-Light) accepted under full load, heavier ones rejected. One device, software limits, no network, no privacy. Saber is not a NIST standard.
[7]	Santos et al., ISDFS 2026	ML-KEM-768 and ML-DSA-65 hybrid TLS for MQTT, emulated ARM64 with 0.2% loss and 30 ms jitter	Connection set-up took 212 ms against 44 ms classical (4.7 times). The cost was computation, not bandwidth. Later publishing was negligible.
[8]	Karthikeyan, GCITC 2023	RSA with Twofish plus NTRUEncrypt, deep-learning key selection, watermarking and blockchain for medical images	Reports a 50 to 75% cut in encryption overhead. Uses NTRU, not a NIST-standardised scheme. No network adaptation.
[9]	Zhukabayeva et al., IEEE Access 2024	Hash-based post-quantum cryptography on Hyperledger Fabric for patient records	Discrete-event simulation of transactions, requests and encryption. No real network.
[10]	Aydeger and Carvalho, IEEE CCNC 2026	Cryptographic agility for UAV swarms, 600 simulated missions	At 50 UAVs, static high security met 7.5% of deadlines while agile and static low met 90%. Security floors are soft and may be dropped under load.
[11]	Ojetunde et al., IEEE CCNC 2025	PQC and PQC-AES hybrid public-key protocols over a wireless file transfer	Kyber1024 and BIKE-L1 matched AES-256 throughput (about 0.14 Mbit/s) for 1 KB files. Static choices.
[12]	Lokesh, NMITCON 2025	Kyber512 inside TLS 1.3 on Raspberry Pi hardware	14.1 ms handshake and 0.36 mJ per handshake on Raspberry Pi 3. Lists crypto agility only as a table entry and describes no run-time controller.
[13]	Rubio Garcia et al., IEEE CAMAD 2023	Triple-hybrid TLS 1.3 with classical, post-quantum and quantum key distribution	About 68% more handshake time than a classical plus post-quantum hybrid.
[14]	Catoni et al., IEEE SMARTCOMP 2025	X25519MLKEM768 and TLS 1.3 session resumption in edge-cloud proxies	Resumption cut connection cost by up to 73%.

Ref	Paper (venue, year)	What it does	Result or limit that matters here
[15]	Rigon et al., IEEE CNS 2025	ML-KEM, BIKE, HQC and FrodoKEM, plain and hybrid, inside QUIC	ML-KEM was the most efficient and hybrids added little. Dilithium matched classical signatures and SPHINCS+ slowed the handshake.
[16]	Reazaei et al., IEEE CSPA 2026	Review of ML-KEM, BIKE and HQC	ML-KEM is the primary choice and HQC the standardised backup. A review with no network test.
[17]	He, Wang and Cai, IEEE IoT J. 2024	Local differential privacy for clustered federated learning (ACS-FL) with adaptive clipping	Tested on image datasets, not health data. The adaptation is per layer inside training.
[18]	Yu et al., IEEE BIBM 2024	Adaptive noise and privacy budget from gradient components in medical federated learning	Accuracy up 3.47% on RSNA-ICH. The budget follows the gradients, not the link.
[19]	Zhang et al., IEEE Healthcom 2023	Lightweight attribute-based encryption with differential privacy for medical IoT data	Cuts computation and communication cost in simulation. Static policy.
[20]	Lyu et al., IEEE Trans. Big Data 2025	Edge-DPSDG: synthetic medical data with budgets set by Shapley value and entropy	Up to 21.29% better accuracy than the state of the art. Budgets are set once per dataset.
[21]	Birajdar et al., IEEE ICAAIC 2025	Dynamic privacy-budget framework with Rényi accounting for medical federated learning	Theory only, with no experiments.

5.3 Reading across the papers

Three groups emerge. The first group, [4, 5, 7, 11, 12, 13, 14, 15, 16], measures post-quantum cost on one fixed configuration at a time. Its most useful lesson for this project comes from [7], which attributes the 4.7-fold handshake penalty to computation rather than bandwidth, and from [14], which shows that resumption removes most of the cost for long-lived sessions. Both support the choice to re-key make-before-break rather than to re-handshake on every change. The second group, [6, 10], does adapt post-quantum parameters at run time, but for a single device or a simulated swarm, and neither adapts privacy nor enforces a hard floor. The third group, [17, 18, 19, 20, 21], adapts or optimises privacy parameters for medical data, but reacts to model training or the dataset, and never to the network or the transport cipher. Papers [8, 9] apply cryptography to healthcare data; one uses a scheme outside the NIST standards and the other evaluates by simulation only.

5.4 Positioning against this work

5.5 Research gap

- **No paper joins post-quantum cryptography and privacy.** Of the 18, none sets a differential-privacy or anonymity parameter together with a post-quantum suite. The healthcare papers [4, 5, 8, 9] protect transport or storage with fixed settings. The privacy

Ref	Healthcare	Post-quantum	Privacy (DP or anonymity)	Adapts at run time	Tested on a network
[4]	yes	yes	no	no	cluster, compute only
[5]	yes	yes	authentication privacy	no	analysis
[6]	no	yes	no	yes, CPU and RAM rules	no
[7]	no	yes	no	no	emulated
[8]	yes	yes (NTRU)	no	no	no
[9]	yes	yes	no	no	simulated
[10]	no	yes	no	yes, soft floors	simulated
[11]	no	yes	no	no	wireless test
[12]	no	yes	no	no	hardware, loss tested
[13]	no	yes	no	no	experimental network
[14]	no	yes	no	no	several conditions
[15]	no	yes	no	no	controlled
[16]	no	yes	no	no	review
[17]	no	no	yes	yes, inside training	no
[18]	yes	no	yes	yes, from gradients	no
[19]	yes	no	yes	no	simulated
[20]	yes	no	yes	once per dataset	no
[21]	yes	no	yes	yes, theory	no
This work	yes	yes	yes	yes, from latency budget	emulated and real shaped link

Table 3: Positioning of this work against the 18 surveyed papers.

papers [17]–[21] never touch the transport cipher.

- **Run-time adaptation is thin.** Only [6] (rule-based, one device) and [10] (UAV simulation) adapt post-quantum settings at run time. Neither is for healthcare, neither adapts privacy, and neither is tested on a shaped link. Adaptive privacy [17, 18, 21] reacts to training, not to network latency, and [21] is theory only. Reference [20] sets its budgets once per dataset, so it does not adapt at run time.
- **Floors are soft or absent.** Reference [10] allows its security floor to drop under load and lists formal floor checks as future work. This project holds raw clinical data at NIST Category 3 or above, and the floor held in every trial.
- **Network evidence is mostly static.** The cost studies [7, 11, 12, 13, 14, 15] measure one fixed configuration at a time. Reference [7] finds that computation, not bandwidth, caused the 4.7-fold handshake penalty, and [14] shows that resumption cuts connection cost by up

to 73%. Both point to long-lived sessions, which fits the make-before-break re-keying used here. They also expose a limit: this project's real-link test does not charge post-quantum computation to latency.

- **Not novel.** Run-time selection of post-quantum settings already exists [6, 10, 24, 25, 26], and so do adaptive or data-driven privacy budgets [18, 20, 21, 27]. This project does not claim either.
- **Claimed gap.** Joint control of transport cryptography and privacy through one latency budget, with a hard floor for raw clinical data, evaluated on emulated and real shaped links. A 2026 survey lists this direction as open [28].

6 Detailed Description

6.1 Background

Healthcare data stays sensitive for decades, so it must survive the arrival of quantum computers. An attacker can record encrypted traffic today and decrypt it later. NIST has standardised the replacements: ML-KEM for key exchange [1] and ML-DSA for signatures [2]. Guidance for the transition recommends hybrid use of classical and post-quantum algorithms [3].

A hybrid combines a classical and a post-quantum primitive so that the result stays secure if either one holds. In this project the key exchange is X25519 plus ML-KEM, and the signature is Ed25519 plus ML-DSA. Post-quantum algorithms are larger and slower than the classical ones they replace. A hospital link carrying ECG streams and clinical records has a real-time budget, so the strongest setting is not always the right one. The project turns that trade-off into a run-time decision.

6.2 Threat model and assumptions

- **Passive quantum adversary.** Records traffic now and attacks the key exchange later. Countered by the hybrid X25519 + ML-KEM exchange.
- **Active network attacker.** Forges, modifies or replays packets. Countered by Ed25519 on every packet plus ML-DSA per epoch over a hash chain, and by AEAD on every record.
- **Curious insider or researcher.** Tries to re-identify patients from exported data. Countered by differential privacy, (K, L)-anonymity, and a researcher export floor that is independently verified.
- **Assumptions.** Endpoints are trusted, the underlying algorithms are secure, and the network is untrusted. The project is software-only, so hardware side channels are out of scope.

6.3 Two traffic domains

Raw patient data is identifiable, so lowering its protection to save time is never acceptable. De-identified research data already carries the privacy transform, so it can use the full range and give up cryptographic strength on a congested link.

Domain	Data	Policy
Clinical	Raw PHI (ECG, EHR)	Tier is max(adaptive tier, 3). Never drops below NIST Category 3.
Research	De-identified data	Full tier range, with privacy transforms applied before export.

Table 4: The two domains and their policies.

6.4 Latency budget model

The controller measures round-trip time (RTT) and CPU load, then computes the time left for cryptography:

$$L = 200 - \mathrm{RTT} - 0.5 \cdot \mathrm{CPU} \tag{1}$$

Here L is the latency budget in ms, RTT is in ms and CPU is a percentage. The constant 200 ms is the end-to-end tolerance of streaming telemetry. A large L means that the link can afford strong cryptography. A small L means that the link is stressed. The CPU term accounts for the time that a loaded gateway needs to execute the cryptography itself.

6.5 Three security levels

Level	Trigger	KEM	Signature	Cipher	DP ε	(K, L)
High (Tier 5)	$L > 80$	ML-KEM-1024	ML-DSA-87	AES-256-GCM	0.1	(5, 3)
Medium (Tier 3)	$40 < L \leq 80$	ML-KEM-768	ML-DSA-65	AES-256-GCM	0.5	(3, 2)
Low (Tier 1)	$L \leq 40$	ML-KEM-512	ML-DSA-44	ChaCha20-Poly1305	1.0	(2, 2)

Table 5: The three tiers bind transport and privacy parameters together.

Level	KEM public key (B)	KEM ciphertext (B)	Sig. public key (B)	Signature (B)
Tier 5	1568	1568	2592	4627
Tier 3	1184	1088	1952	3309
Tier 1	800	768	1312	2420
Classical	32 (X25519)	32 (X25519)	32 (Ed25519)	64 (Ed25519)

Table 6: Object sizes grow with the level, which is why the lower levels help on thin links.

A smaller differential-privacy ε means stronger privacy, and a larger (K, L) means stronger anonymity. The tiers therefore couple strong cryptography with strong privacy on a healthy link and relax both together on a stressed one, with research data only. A relaxation of privacy never applies to raw clinical data, which is held at Tier 3 or above.

6.6 Governance requirements

Healthcare deployments need accountability as well as confidentiality. The system therefore defines four roles (administrator, auditor, researcher and operator), issues JSON Web Tokens for access, writes every action to a hash-chained HMAC audit log, and exposes a tamper lab for demonstrating that modified traffic is rejected.

7 Proposed Work and Design

7.1 Architecture

One controller decides the security level, and both the secure channel and the privacy engines follow that decision (Figure 1).

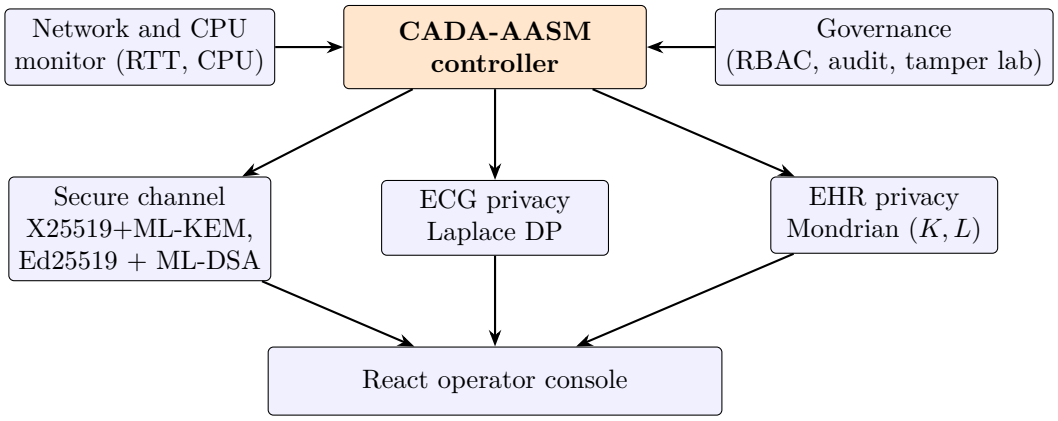


Figure 1: System architecture: five components and one controller. The monitor feeds the controller, which sets the tier for the channel and both privacy engines.

7.2 Control algorithm

The controller runs once per cycle. It reads RTT and CPU load, computes the budget L, maps L to a tier, and applies a persistence filter so that one noisy sample cannot trigger a switch.

$$\text{tier}(L) = \begin{cases} 5 & L > 80 \\ 3 & 40 < L \leq 80 \\ 1 & L \leq 40 \end{cases} \qquad \text{tier}_{\text{clinical}} = \max\big(\text{tier}(L),\, 3\big) \tag{2}$$

1. Measure RTT and CPU, and compute $L = 200 - \text{RTT} - 0.5\,\text{CPU}$.
2. Map L to a candidate tier by the thresholds above.
3. If the candidate equals the current tier, reset the counter and stop.
4. Otherwise count how many consecutive cycles have proposed the same candidate.
5. When the candidate has been proposed for **3 consecutive cycles**, adopt it and start a make-before-break re-key.
6. Compute the clinical tier as $\max(\text{tier}, 3)$, so that raw patient data never falls below NIST Category 3. The research domain uses the tier directly.

The persistence filter trades a delay of up to three cycles in reacting for far fewer tier changes. In the emulated evaluation it cut tier changes from 6.9 to 2.4 per run (a 65% reduction), and the number of handshakes from 10.9 to 6.4.

7.3 Secure channel

1. **Hybrid key exchange.** X25519 and the ML-KEM variant of the current tier each produce a shared secret.
2. **Key derivation.** HKDF combines both secrets, salted with the transcript hash, so the session key depends on the whole handshake.
3. **Authentication.** Ed25519 and the tier's ML-DSA variant sign the handshake.
4. **Data protection.** AES-256-GCM, or ChaCha20-Poly1305 at Tier 1, protects each record.
5. **Re-keying.** When the tier changes, the new session is established before the old one is retired (make-before-break), so traffic is never interrupted.
6. **Batch signing.** Each packet carries an Ed25519 signature. ML-DSA signs once per epoch of 8 packets over a hash chain of those packets. Setting the epoch length to 1 gives the per-packet baseline.

Batch signing is the main bandwidth saver. In the evaluation the wire size per clinical exchange fell from about 5.0 kB to 0.9 kB, with 1.1% SLA violations.

7.4 Privacy engines

- **ECG streams.** Laplace noise with sensitivity $\Delta = 0.5\,\text{mV}$ and the tier's ε [29]. A smaller ε gives stronger privacy: 0.1 at Tier 5, 0.5 at Tier 3 and 1.0 at Tier 1.
- **EHR records.** Mondrian multidimensional (K, L)-anonymity [30, 31, 32] with K and L from the tier table. A researcher export is refused unless $K \geq 5$ and $L \geq 3$, and a separate verifier independently confirms the result before release.

7.5 Governance and interface

- **Roles.** JWT-based access for administrator, auditor, researcher and operator.
- **Audit.** Every action is written to a hash-chained HMAC log, so any edit to history is detectable.
- **Tamper lab.** Lets an operator flip bits in transit and watch the channel reject the packet.
- **Dashboard.** A React 18 interface shows the live tier, budget, charts, privacy settings and audit trail.

7.6 Implementation

The backend is Flask with Socket.IO. The frontend is React 18 with Vite, Recharts and socket.io-client. Cryptography uses the pure-Python `kyber-py` and `dilithium-py` libraries, plus a native `liboqs` 0.16.0 benchmark for realistic timings. The suite has 65 `pytest` tests, and a Linux network-namespace testbed validates behaviour on a real shaped link.

7.7 Evaluation

Emulated link. 30 paired trials with staged calm, moderate, heavy, severe and calm periods.

Comparison	Clinical violations	Research violations	Significance
Static Tier 5	7.4%	6.9%	baseline
Adaptive (CADA-AASM)	4.1%	2.9%	$p < 10^{-4}$ vs Tier 5
Adaptive vs static Tier 3	not different	not different	$p = 0.16$ and 0.058

Table 7: Emulated evaluation, 30 paired trials.

The persistence filter cut tier changes from 6.9 to 2.4 per run and handshakes from 10.9 to 6.4. In the bandwidth and loss sweep the gain reached about 45 percentage points at 1 Mbit/s. The clinical floor held in every trial.

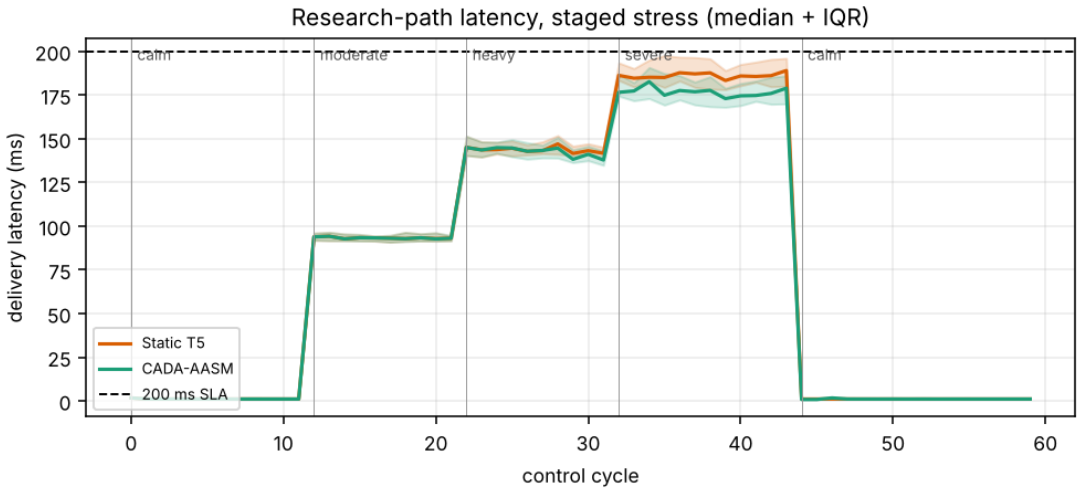


Figure 2: Research-path latency under staged stress (median, IQR over 30 trials).

Real shaped link. Linux network namespaces, `veth` pairs, kernel token-bucket shaping and real TCP. Propagation delay is a stated receiver hold, random loss is not modelled and post-quantum computation is not charged to latency.

Constant severe stress, clinical violations	5 Mbit/s	2 Mbit/s	1 Mbit/s	0.5 Mbit/s
Adaptive	0%	0%	3.3%	13.3%
Static Tier 5	0%	0%	0%	100%
Static Tier 3	0%	0%	0%	0%

Table 8: Real-link validation. Part B used 4 repetitions, so the smallest possible Wilcoxon p is 0.125 and the results are descriptive.

Under staged stress no configuration had a violation. Mean clinical p95 latency was 159.2 ms for adaptive, 164.8 ms for static Tier 5 and 159.2 ms for static Tier 3. The minimum clinical tier was

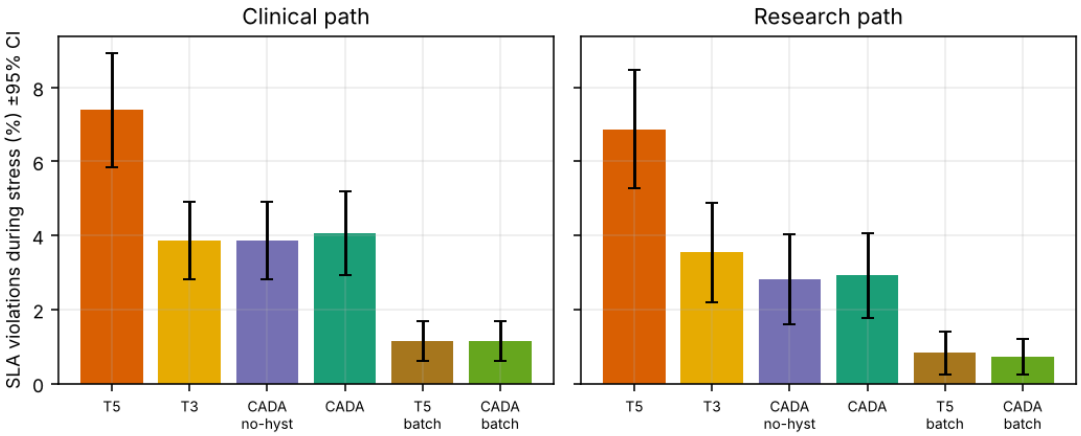


Figure 3: SLA violations during stress with 95% confidence intervals.

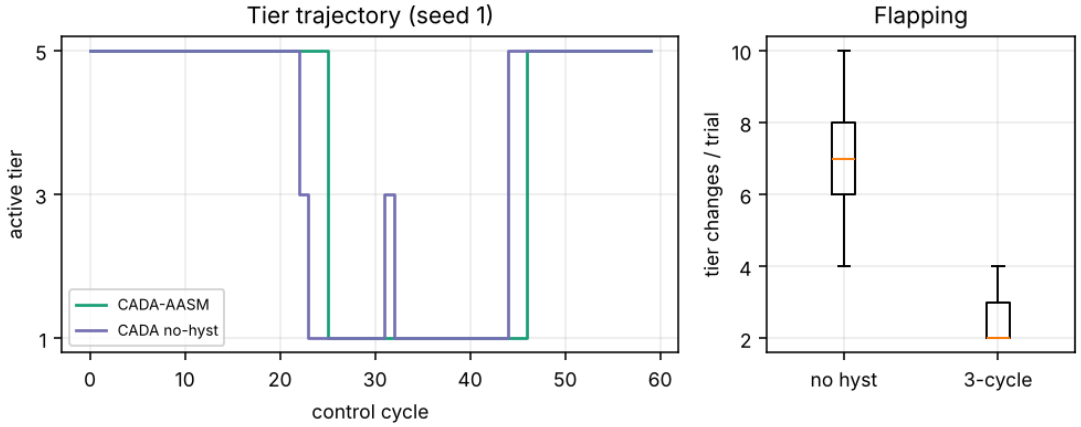


Figure 4: Tier trajectory and flapping with and without the persistence filter.

3 in every adaptive trial. Wire bytes per clinical exchange averaged 5,020 for Tier 5, 3,702 for Tier 3 and about 932 for Tier 5 with batch signing.

Native cryptographic cost. Median times measured with `liboqs` 0.16.0:

7.8 Findings, limitations and future work

Findings. The emulated gain over static Tier 5 is not reproduced on a real link at 2 Mbit/s and above, where every configuration meets the budget. Adaptation matters only on very thin links, and a well-chosen static Tier 3 matches or beats the adaptive controller on latency. The defensible contribution is therefore the joint, safe adjustment of cryptography and privacy under a compliance floor, not a latency win. This is consistent with [7], which found that computation rather than bandwidth dominated the post-quantum penalty.

Limitations.

- Propagation delay is a stated receiver hold, and random loss is not modelled on the real link.
- Part B of the real-link test has few repetitions (4), so its statistics are descriptive.

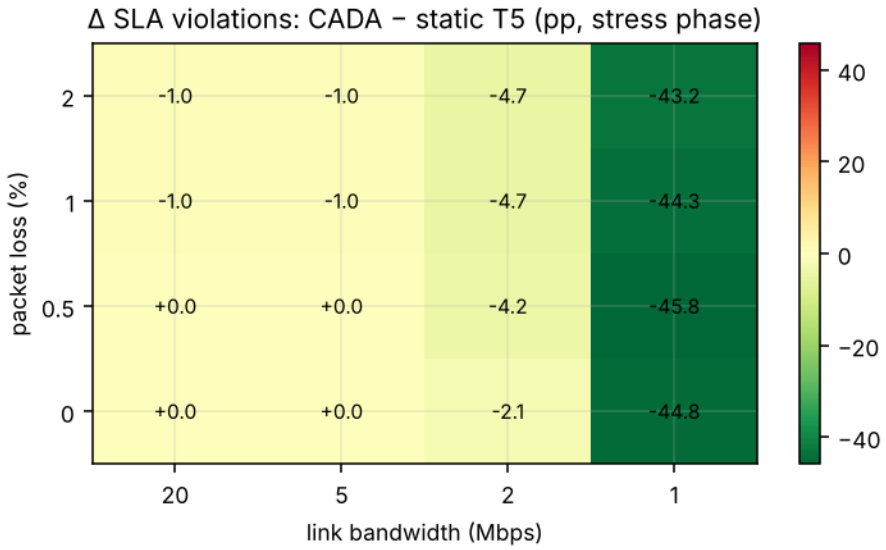


Figure 5: Where steering helps: bandwidth × loss sweep.

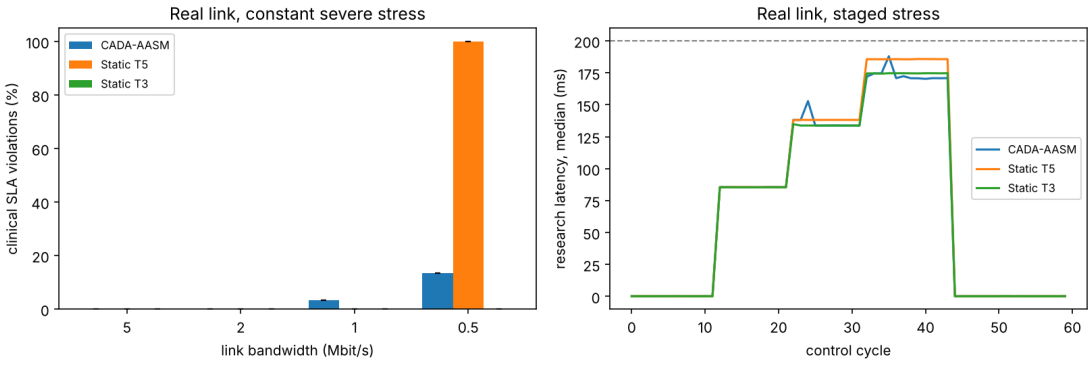


Figure 6: Real-link validation (Linux namespaces and kernel token-bucket shaping). Left: clinical SLA violations under constant severe stress against bandwidth. Right: median research latency under staged stress.

- Pure-Python cryptography is much slower than native, and native `liboqs` is not yet used inside the channel.
- Post-quantum computation is not charged to latency in the real-link test, which [7] suggests is where much of the real cost lies.
- The literature survey is a single-reviewer, IEEE-only search and is not a systematic review.

Future work. Multi-host virtual-machine testing with kernel delay and loss; native `liboqs` inside the channel with computation charged to latency; and a learned controller compared against the rule-based one.

Operation	**Median time (ms)**
ML-KEM-768 keygen / encaps / decaps	0.0184 / 0.0136 / 0.0153
ML-DSA-44 sign	0.0613
ML-DSA-65 sign / verify	0.0872 / 0.0321
ML-DSA-87 sign	0.1077
Pure-Python ML-DSA signing in this project	about 94

Table 9: Native cost is orders of magnitude below the 200 ms budget, whereas the pure-Python implementation is not.

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